Project: Worked examples
Title: Column buckling (Perry–Robertson)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Compressive strength and resistance of a pin-ended strut of a given effective length, from Euler's critical stress reduced for initial imperfection by the Perry–Robertson formula, with the Robertson constant chosen by strut curve; utilisation against an applied axial load.

WHAT IT CALCULATES	RELATION
Radius of gyration and slenderness	$r = \sqrt{(I/A)}$; $\lambda = L_E/r$
Euler buckling stress and load	$p_E = \pi^2 E/\lambda^2$; $P_E = p_E \cdot A$
Limiting slenderness, below which the strut is not reduced for buckling	$\lambda_0 = 0.2 \cdot \sqrt{(\pi^2 E/p_y)}$
Perry factor (Robertson constant a = 2.0, 3.5, 5.5, 8.0 for curves a–d)	$\eta = \max(0.001 \cdot a \cdot (\lambda - \lambda_0), 0)$
Compressive strength by the Perry–Robertson formula	$\varphi = (p_y + (\eta + 1) \cdot p_E)/2$; $p_c = p_E \cdot p_y/(\varphi + \sqrt{(\varphi^2 - p_E \cdot p_y)})$
Compression resistance and utilisation	$P_c = p_c \cdot A$; $u = P/P_c$

Basis: Compressive strength (Perry strut formula)[1](Annex C); Maximum slenderness[1](4.7.3.2); Allocation of strut curve[1](Table 23).

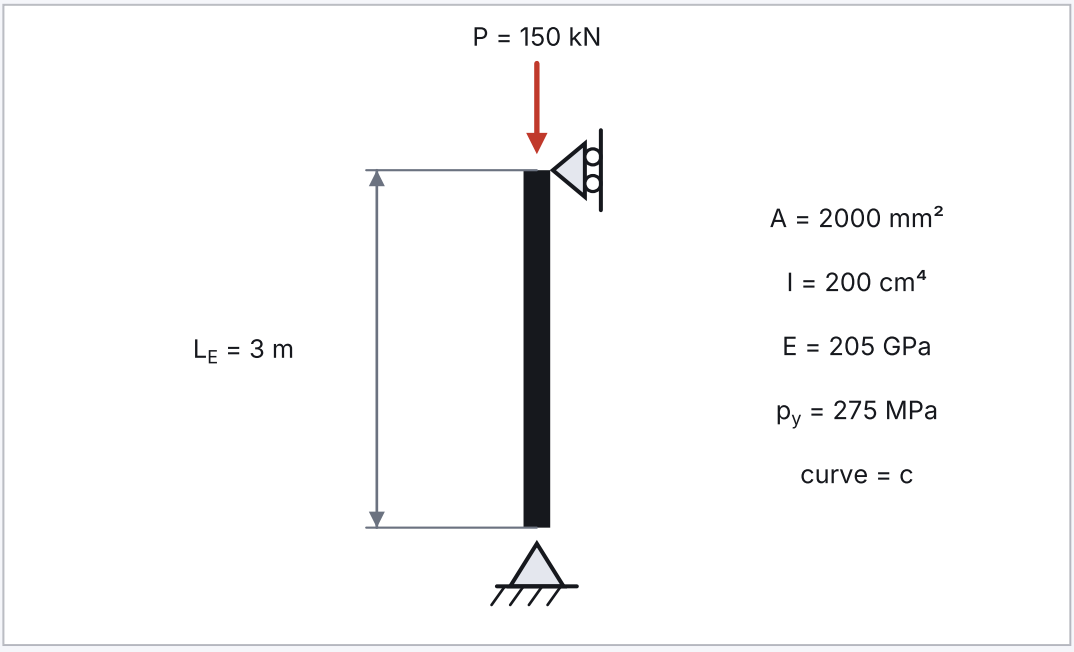


Figure 1 — A pin-ended strut under axial compression, effective length L_E. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L_E	Effective length	3	m	drawing: DRG-ST-410 rev B	0.1 … 50 m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
A	Cross-section area	2000	mm²	spec: SPEC-ST-02 section schedule	10 … 1000000 mm²
I	Second moment of area about the weaker axis	200	cm⁴	spec: SPEC-ST-02 section schedule	0.01 … 10000000 cm⁴
E	Modulus of elasticity	205	GPa	spec: BS 5950-1 3.1.3: E = 205 GPa	50 … 450 GPa
p_y	Design strength (yield)	275	MPa	spec: SPEC-ST-02: S275, t ≤ 16 mm	100 … 1000 MPa
curve	Strut curve (selects the Robertson constant)	c		spec: BS 5950-1 Table 23: welded box, curve c	any
P	Applied axial load	150	kN	assumption: Factored axial load from the frame analysis	0 … 100000 kN

3. Assumptions

A1. Pin-ended, prismatic strut under axial load only; buckling about the axis of the given second moment. (Effective length carries the end restraint.)
A2. Initial imperfection represented by the Perry factor $\eta = 0.001 \cdot a \cdot (\lambda - \lambda_0)$, with a from the strut curve. (Robertson's fit to strut tests.)
A3. P (Applied axial load) is taken as 150 kN. (Factored axial load from the frame analysis)

4. Working

[8] $a_R = 5.5$ [1](Annex C)
$= 5.5$
Robertson constant for strut curve c
[9] $r = \mathrm{sqrt}(I / A)$
$= \mathrm{sqrt}(200\ \mathrm{cm^4} / 2000\ \mathrm{mm^2})$
$= 31.623\ \mathrm{mm}$
Radius of gyration
[10] $\lambda = L_E / r$
$= 3\ \mathrm{m} / 31.623\ \mathrm{mm}$
$= 94.868$
Slenderness
[11] $p_E = \mathrm{pi}^2 * E / \lambda^2$ [1](Annex C)
$= \pi^2 \times 205\ \mathrm{GPa} / 94.868^2$
$= 224.81\ \mathrm{MPa}$
Euler critical stress
[12] $P_E = p_E * A$
$= 224.81\ \mathrm{MPa} \times 2000\ \mathrm{mm^2}$
$= 449.62\ \mathrm{kN}$
Euler critical load
[13] $\lambda_0 = 0.2 * \mathrm{sqrt}(\mathrm{pi}^2 * E / p_y)$ [1](Annex C)
$= 0.2 \times \mathrm{sqrt}(\pi^2 \times 205\ \mathrm{GPa} / 275\ \mathrm{MPa})$
$= 17.155$
Limiting slenderness
[14] $\eta = \max(0.001 * a_R * (\lambda - \lambda_0), 0)$ [1](Annex C)
$= \max(0.001 \times 5.5 \times (94.868 - 17.155), 0)$
$= 0.42742$
Perry factor (initial imperfection)

[15] $\varphi = (p_y + (\eta + 1) * p_E) / 2$
$= (275\ \text{MPa} + (0.42742 + 1) \times 224.81\ \text{MPa}) / 2$
$= 297.95\ \text{MPa}$
Perry–Robertson φ

[16] $p_c = p_E * p_y / (\varphi + \sqrt{\varphi^2 - p_E * p_y})$ [1](Annex C)
$= 224.81\ \text{MPa} \times 275\ \text{MPa} / (297.95\ \text{MPa} + \sqrt{(297.95\ \text{MPa})^2 - 224.81\ \text{MPa} \times 275\ \text{MPa}})$
$= 133.78\ \text{MPa}$
Compressive strength

[17] $P_c = p_c * A$
$= 133.78\ \text{MPa} \times 2000\ \text{mm}^2$
$= 267.56\ \text{kN}$
Compression resistance

[18] $u = P / P_c$
$= 150\ \text{kN} / 267.56\ \text{kN}$
$= 0.56062$
Utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
r	31.623 mm	Radius of gyration
λ	94.868	Slenderness
p_E	224.81 MPa	Euler stress
P_E	449.62 kN	Euler load
λ_0	17.155	Limiting slenderness
η	0.42742	Perry factor
p_c	133.78 MPa	Compressive strength
P_c	267.56 kN	Compression resistance
u	0.56062	Utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Axial load against compression resistance	u = 0.56062	u ≤ 1	0.561	[1](Annex C)

7. Applicability

No warnings.

8. Verification

Template struct.column-buckling@1.0.1 Content hash b92d254599e2302a Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Structural Use of Steelwork in Building—Part 1: Code of Practice for Design—Rolled and Welded Sections. BS 5950-1:2000. BSI; 2000.